4MA1_1H_ch1_numbers_and_number_system

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1. 4MA1/1H_Winter_2024_Q21 — 1 point

$$T = \frac{x^2 + y^2}{w}$$

$x = 28.4$ correct to 1 decimal place.

$y = 17$ correct to 2 significant figures.

$w = 90$ correct to the nearest 5

Calculate the upper bound for the value of T

Give your answer correct to 3 significant figures.
Show your working clearly.

3 marks

2. 4MA1/1H_Winter_2024_Q17 — 1 point

(a) Express $\sqrt{675}$ in the form $n\sqrt{27}$ where n is a positive integer.

....

(1)

(b) Show that $\dfrac{5-\sqrt{2}}{\sqrt{2}-1}$ can be written in the form $a + b\sqrt{2}$ where a and b are integers.

(3)

3. 4MA1/1H_Winter_2024_Q9 1 point

(a) Write 8.4×10^{-5} as an ordinary number.

(1)

(b) Work out $(6.5 \times 10^{-40}) \times (8 \times 10^{185})$
Give your answer in standard form.

(2)

4. 4MA1/1H_Winter_2024_Q7 1 point

Shop **A** and Shop **B** have offers for buying the same type of laptop.

The normal price of the laptop in Shop **A** is different to the normal price of the laptop in Shop **B**

Shop A	**Shop B**
Our normal price £475	Get 15% off our normal price
Get 16% off our normal price	Only pay £408

Which shop gives more money off their normal price?
Show your working clearly.

4 marks

5. 4MA1/1H_Winter_2024_Q6 1 point

$A = 2^3 \times 5^4 \times 7 \times 11$

$B = 2^2 \times 5^2 \times 7^2$

$C = 2^2 \times 5^3 \times 7^4$

Find the highest common factor (HCF) of A, B and C
Write your answer as a product of prime factors.

2 marks

6. 4MA1/1HR_Summer_2024_Q17 1 point

(a) $\left(\sqrt[4]{k^{12}}\right)^5 = k^n$

Find the value of n

(1)

(b) Express $\dfrac{7}{2-\sqrt{3}}$ in the form $\sqrt{c} + d$ where c and d are integers.

Show your working clearly.

(3)

7. 4MA1/1HR_Summer_2024_Q11 1 point

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

(a) Write down the highest common factor (HCF) of $5A$ and $2B$
Give your answer as a product of prime factors.

(2)

$A = 2^5 \times 5 \times 7^2$

$B = 2^3 \times 5^3 \times 7^4$

(b) Work out the value of $(AB)^2$
Give your answer as a product of prime factors.

(2)

8. 4MA1/1HR_Summer_2024_Q5 1 point

Slavomir invests 5200 euros in a savings account for 4 years.
He gets 2.5% per year compound interest.

Work out how much money Slavomir will have in the savings account at the end of 4 years.
Give your answer correct to the nearest euro.

.... euros

(3)

9. 4MA1/1HR_Summer_2024_Q4

1 point

Show that $2\frac{1}{3} \div 5\frac{1}{4} = \frac{4}{9}$

(3)

10. 4MA1/1HR_Summer_2024_Q3

1 point

Norberto sells white loaves of bread and brown loaves of bread.

He sells a total of 200 loaves such that

the number of white loaves sold : the number of brown loaves sold = 3 : 2

Norberto sells the white loaves for £1.50 each.
He sells the brown loaves for £1.75 each.

40% of the price of a white loaf is profit.
60% of the price of a brown loaf is profit.

Work out Norberto's total profit when he sells all 200 loaves.

£..............................

(5)

11. 4MA1/1HR_Summer_2024_Q1

1 point

Here are six cards.

Five of the cards have a number written on them.

16	15	3	2	9	

Work out the number that should be written on the last card so that the mean of the six numbers will be 11

(3)

12. 4MA1/1H_Summer_2024_Q17

1 point

Show that $\frac{1+\sqrt{5}}{3-\sqrt{5}}$ can be written in the form $a+\sqrt{b}$ where a and b are integers.

Show each stage of your working clearly.

(3)

13. 4MA1/1H_Summer_2024_Q14 1 point

The combined savings of Abel and Bahira are 15 435 dinars.

The savings of Bahira are 45% more than the savings of Abel.

The savings of Bahira are $\frac{3}{2}$ times the savings of Chanda.

Work out the savings of Chanda.

.... dinars

(5)

14. 4MA1/1H_Summer_2024_Q9 1 point

The table gives the amount of rice produced by each of two countries in 2020

Country	**Amount of rice (tonnes)**
Indonesia	3.5×10^7
Argentina	8.2×10^5

(a) Write 3.5×10^7 as an ordinary number.

(1)

In 2020, Japan produced 6 780 000 more tonnes of rice than Argentina.

(b) Work out the amount of rice Japan produced in 2020
Give your answer in standard form.

.... tonnes

(2)

15. 4MA1/1H_Summer_2024_Q4 1 point

Ava records the number of kilometres she drives each month.

In April, Ava drove 943 kilometres.
This is 15% more than the number of kilometres she drove in March.

Work out the number of kilometres Ava drove in March.

.... kilometres

(3)

16. 4MA1/1H_Summer_2024_Q3 1 point

Find the highest common factor (HCF) of 72 and 108
Show your working clearly.

(2)

17. 4MA1/1HR_Winter_2023_Q14 1 point

$T = \frac{p}{r}$

$p = 0.51$ correct to 2 significant figures.
$r = 6.3$ correct to 2 significant figures.

Work out the upper bound for the value of T
Show your working clearly.

....

2 marks

18. 4MA1/1HR_Winter_2023_Q7

1 point

(a) Write 9.32×10^{-5} as an ordinary number.

....

(1)

(b) Work out $3 \times 10^5 - 6 \times 10^4$

Give your answer in standard form.

....

(2)

(c) Work out $(3 \times 10^{55}) \times (6 \times 10^{65})$

Give your answer in standard form.

....

(2)

19. 4MA1/1HR_Winter_2023_Q5

1 point

C grams of chocolate is shared in the ratios $2:5:8$
The difference between the largest share and the smallest share is 390 grams.

Work out the value of C

$C =$ ....

3 marks

20. 4MA1/1HR_Winter_2023_Q4

1 point

Divya and Yuan each pay for a holiday at a special offer price.

Divya's holiday	**Yuan's holiday**
Normal price: \$1600	Normal price: \$1400
Special offer: 16% off the normal price	Special offer: k% off the normal price

The amount that Divya pays is the same as the amount that Yuan pays.

Work out the value of k

k = ...

4 marks

21. 4MA1/1H_Winter_2023_Q19 1 point

The acceleration, a, of an object is given by

$$a = \frac{v - u}{t}$$

where

$v = 45.23$ correct to 2 decimal places
$u = 5.12$ correct to 2 decimal places
$t = 8.5$ correct to 2 significant figures

By considering bounds, work out the value of a to a suitable degree of accuracy.
Show your working clearly and give a reason for your answer.

$a =$ ....

5 marks

22. 4MA1/1H_Winter_2023_Q9 1 point

(a) Write 6.25×10^{-4} as an ordinary number.

....

(1)

(b) Work out $(2.4 \times 10^{12}) \div (9.6 \times 10^{4})$
Give your answer in standard form.

....

(2)

Shane bought a car.

The amount Shane paid for the car was \$32 000

Theresa also bought a car.
To pay for this car, Theresa paid a deposit of \$18 000 together with 14 monthly payments of \$1160

Theresa paid more for her car than Shane paid for his car.

(a) Work out how much more Theresa paid as a percentage of the amount Shane paid.

.... %
(4)

Kylie bought a van.

After 1 year, the value of the van was \$39 865
During this year, the value of the van decreased by 15%

(b) Work out the value of the van when Kylie bought it.

\$ ....
(3)

In a box, there are only green sweets, orange sweets and yellow sweets.

There are 280 sweets in the box so that

the number of green sweets : the number of orange sweets = 2 : 3

and

the number of orange sweets : the number of yellow sweets = 1 : 5

Work out how many green sweets there are in the box.

....

3 marks

25. 4MA1/1HR_Summer_2023_Q18 1 point

Diego builds a fence using fence panels.

The total length of the fence is 50 metres, correct to the nearest 5 metres.
The length of each fence panel is 1.3 metres, correct to the nearest 10 cm.

The cost of each fence panel is £8.65

Diego may only buy complete fence panels.
Diego only pays for the number of panels he needs to build the fence.

Work out the greatest difference in the possible amounts that Diego could pay to build the fence.
Show your working clearly.

£..............................

4 marks

26. 4MA1/1HR_Summer_2023_Q9 1 point

Kazi buys a car for 700 000 taka.
The value of the car depreciates by 12% each year.

Work out the value of the car at the end of 3 years.
Give your answer correct to the nearest taka.

.... taka

3 marks

27. 4MA1/1HR_Summer_2023_Q8 1 point

(a) Write 5.6×10^{-3} as an ordinary number.

....

(1)

(b) Work out $\dfrac{6 \times 10^{3}}{2.1 \times 10^{-4} + 9 \times 10^{-5}}$

Give your answer in standard form.

....

(2)

28. 4MA1/1HR_Summer_2023_Q1 1 point

Write 2250 as a product of powers of its prime factors.
Show your working clearly.

....

3 marks

29. 4MA1/1H_Summer_2023_Q15 1 point

(a) $\sqrt{2} \div \dfrac{8^3}{16^{\frac{3}{2}}} = 2^n$

Work out the value of n
Show your working clearly.

$n =$ ....

(3)

(b) Find 4% of 4.5×10^{157}
Give your answer in standard form.

....

(3)

30. 4MA1/1H_Summer_2023_Q8 1 point

(a) Write 300 as a product of its prime factors.
Show your working clearly.

....

(2)

$A = 2 \times 2 \times 2 \times 3 \times 3 \times 5$

$B = 2 \times 2 \times 3 \times 3 \times 3 \times 5$

(b) Find the lowest common multiple (LCM) of $5A$ and $7B$
Show your working clearly.

....

(2)

31. 4MA1/1H_Summer_2023_Q3 1 point

Nanette buys 60 notebooks for a total cost of 780 dirhams.

Nanette sells 70% of the notebooks for 22 dirhams each.
She sells the remaining notebooks for 19 dirhams each.

Work out Nanette's percentage profit.
Give your answer correct to 3 significant figures.

....%

4 marks

32. 4MA1/1H_Summer_2023_Q1 1 point

Last season, the number of goals scored by Arjun, by Simon and by Kath for their football team were in the ratios $2 : 5 : 8$

Simon scored 12 more goals than Arjun.

Work out the number of goals scored by Kath.

....

3 marks

33. 4MA1/1HR_Winter_2022_Q18 1 point

The three solids **A**, **B** and **C** are similar such that

the surface area of **A** : the surface area of **B** = 4 : 9

and

the volume of **B** : the volume of **C** = 125 : 343

Work out the ratio

the height of **A** : the height of **C**

Give your ratio in its simplest form.

....

4 marks

34. 4MA1/1HR_Winter_2022_Q10

1 point

Jonty has a storage container in the shape of a cuboid, as shown in the diagram.

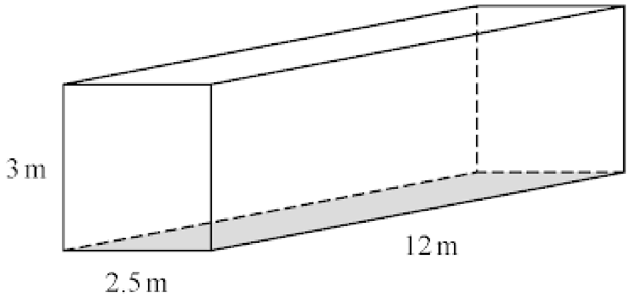


Diagram **NOT** accurately drawn

Jonty is going to paint the outside of his storage container, apart from the base which is shown shaded in the diagram.
He needs enough paint to cover the four sides and the top.

Each tin of paint covers an area of $15\,m^2$

The cost of each tin of paint recently increased by 10%
After the increase, the cost of each tin of paint is £26.95

Jonty says

"**Before** the increase, I could have bought enough paint for less than £200"

Show that Jonty is correct.
Show your working clearly.

35. 4MA1/1HR_Winter_2022_Q8 1 point

(a) Write 5×10^4 as an ordinary number.

....

(1)

(b) Write 0.000 06 in standard form.

....

(1)

(c) Work out $(4 \times 10^{512}) \div (1.6 \times 10^{700})$
Give your answer in standard form.

....

(2)

36. 4MA1/1HR_Winter_2022_Q7 1 point

Milly went on a car journey.
She travelled from Anesey to Breigh to Clando and then to Duckbridge.

For Anesey to Breigh, Milly drove the 245 km in 2.5 hours.
For Breigh to Clando, Milly drove the 220 km at an average speed of 80 km/h
For Clando to Duckbridge, Milly drove at an average speed of 72 km/h in 50 minutes.

Work out Milly's average speed, in km/h, for the journey from Anesey to Duckbridge.
Give your answer correct to one decimal place.

.... km/h

4 marks

(a) Work out the lowest common multiple (LCM) of 36 and 120

....
(2)

$A = 5^2 \times 7^4 \times 11^p$
$B = 5^m \times 7^{n-5} \times 11$

m, n and p are integers such that
$m > 2$
$n > 10$
$p > 1$

(b) Find the highest common factor (HCF) of A and B
Give your answer as a product of powers of its prime factors.

....
(2)

38. 4MA1/1HR_Winter_2022_Q1 1 point

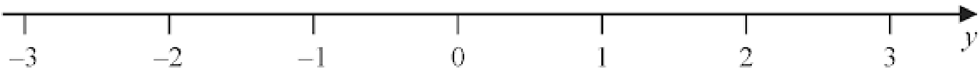
n is an integer.

(a) Write down all the values of n such that $-2 \leqslant n < 3$

....

(2)

(b) On the number line, represent the inequality $y \leqslant 1$



(1)

39. 4MA1/1H_Winter_2022_Q18 1 point

$$A = w - \frac{x^2}{y}$$

$w = 3.45$ correct to 2 decimal places.
$x = 1.9$ correct to 1 decimal place.
$y = 5$ correct to the nearest whole number.

Work out the lower bound of the value of A
Show your working clearly.

....

3 marks

Himari invests 200 000 yen for 3 years in a savings account paying compound interest.

The rate of interest is 1.8% for the first year and x% for each of the second year and the third year.

The value of the investment at the end of the third year is 209 754 yen.

Work out the value of x
Give your answer correct to one decimal place.

$x =$ ....

3 marks

41. 4MA1/1H_Winter_2022_Q8 1 point

$a = 4.2 \times 10^{-24}$ $\qquad$ $b = 3 \times 10^{145}$

Work out the value of $a \times b$
Give your answer in standard form.

....

2 marks

Sarah makes and sells mugs.

One day she makes 150 mugs.
Her total cost for making these mugs is £1140

Of these mugs

$\frac{2}{5}$ are small mugs

32% are medium mugs

and the rest are large mugs

Here is Sarah's price list for selling each mug.

MUGS	
Small	**£8.50**
Medium	**£11.20**
Large	**£14.20**

Sarah sells all 150 mugs.

Work out her percentage profit.
Give your answer correct to the nearest whole number.

....%

5 marks

43. 4MA1/1H_Winter_2022_Q2 1 point

Danil, Gabriel and Hadley share some money in the ratios 3 : 5 : 9
The difference between the amount of money that Gabriel receives and the amount of money that Hadley receives is 196 euros.

Work out the amount of money that Danil receives.

.... euros

3 marks

44. 4MA1/1HR_Summer_2022_Q21

1 point

Express $\dfrac{3+\sqrt{8}}{\left(\sqrt{2}-1\right)^2}$ in the form $p+\sqrt{q}$ where p and q are integers.

Show each stage of your working clearly.

....

4 marks

$P = 3^3 \times 5^2 \times 7$
$Q = 3^2 \times 5 \times 7^2$

(a) Write down the highest common factor (HCF) of P and Q

....
(1)

$P = 3^3 \times 5^2 \times 7$
$Q = 3^2 \times 5 \times 7^2$

(b) Work out the value of $P^3 \times Q$
Give your answer in the form $3^x \times 5^y \times 7^z$ where x, y and z are positive integers.

....
(2)

46. 4MA1/1HR_Summer_2022_Q8 1 point

Jane bought a new car for \$18 000
The car depreciates in value by 15% each year.

Work out the value of the car at the end of 4 years.
Give your answer correct to the nearest \$

\$...

3 marks

47. 4MA1/1HR_Summer_2022_Q6

1 point

Show that $2\frac{2}{3} + 3\frac{3}{4} = 6\frac{5}{12}$

3 marks

48. 4MA1/1HR_Summer_2022_Q4 1 point

The language department of a college has 180 students.
Each student studies exactly one of French, German, Italian or Spanish.

15 students study French.
45% of the students study German.

Express the percentage of students studying Italian or Spanish as a percentage of those studying French or German.

....%

4 marks

49. 4MA1/1HR_Summer_2022_Q3

1 point

Here are five cards.

Each card has a number written on it.

15	7	–2	23	x

The mean of the five numbers is 12

Work out the value of x

$x =$ ....

3 marks

50. 4MA1/1H_Summer_2022_Q16 1 point

Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - \left(\sqrt{2}\right)^5 = 2\sqrt{32} + 12$

Show your working clearly.

3 marks

51. 4MA1/1H_Summer_2022_Q13 1 point

Antoine is going on holiday.
He makes 3 separate payments to cover the total cost of his holiday.

The following table shows how much money Antoine pays to the holiday company.

Payment	**Amount paid**
Payment 1	$\frac{3}{8}$ of the total cost
Payment 2	45% of the total cost
Payment 3	$406

Work out how much Antoine has to pay for Payment 2

$ ....

5 marks

52. 4MA1/1H_Summer_2022_Q8 1 point

(a) Write $0.000\,089$ in standard form.

....

(1)

(b) Write 8.34×10^4 as an ordinary number.

....

(1)

Asha bought an apartment.

The table gives information about the value of apartments, in euros, and the annual service charge band.

Value (x euros)	**Service charge band**
$x \geqslant 700\,000$	A
$600\,000 \leqslant x < 700\,000$	B
$500\,000 \leqslant x < 600\,000$	C
$400\,000 \leqslant x < 500\,000$	D
$0 < x < 400\,000$	E

In 2021, the value of Asha's apartment was 634 400 euros.

The value of Asha's apartment had increased by 4% from its value in 2020

(a) Has the annual service charge band changed for Asha's apartment? Show your working clearly.

(3)

Pam bought a boat.

In each year after Pam bought the boat, the value of the boat depreciated by 15%

(b) Work out the total percentage by which the value of the boat had depreciated by the end of the second year after Pam bought the boat.

.... %

(3)

54. 4MA1/1H_Summer_2022_Q3

1 point

(a) Find the highest common factor (HCF) of 56 and 84
Show your working clearly.

....
(2)

(b) Find the lowest common multiple (LCM) of 60 and 72
Show your working clearly.

....
(2)

55. 4MA1/1HR_Winter_2021_Q19 1 point

$$k = \frac{t}{a - h}$$

$t = 14$ correct to 2 significant figures
$a = 7.8$ correct to 2 significant figures
$h = 3.4$ correct to 2 significant figures

Work out the lower bound for the value of k.
Show your working clearly.

....

(Total for Question is 3 marks)

56. 4MA1/1HR_Winter_2021_Q15 1 point

(a) Use algebra to show that $4.\dot{5}\dot{7} = 4\frac{19}{33}$

(2)

(b) Show that $\frac{2}{6 - 3\sqrt{2}}$ can be written in the form $\frac{a + \sqrt{a}}{b}$

where a and b are integers.
Show your working clearly.

(3)

The diagram shows one face of a wall.
This face is in the shape of a pentagon with exactly one line of symmetry.

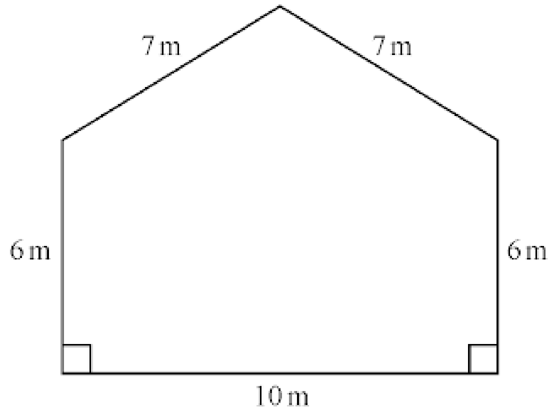


Diagram **NOT** accurately drawn

Omondi is going to paint this face of the wall once.
He has to buy all the paint that he needs to use.

The paint in each tin of paint Omondi is going to buy will cover $16\,m^2$ of the face of the wall.

Work out the least number of tins of paint Omondi will need to buy.
Show your working clearly.

....

(Total for Question is 5 marks)

58. 4MA1/1HR_Winter_2021_Q9 1 point

$A = 2^8 \times 3^5 \times 11^4 \qquad B = 2^6 \times 3 \times 11^8$

(a) Find the highest common factor (HCF) of A and B.

....

(2)

(b) Find the lowest common multiple (LCM) of $2A$ and $3B$.
Give the LCM as a product of powers of its prime factors.

....

(2)

59. 4MA1/1HR_Winter_2021_Q7

1 point

Kuro invests 50 000 yen for 3 years in a savings account.
She gets 2.4% per year compound interest.

Work out how much money Kuro will have in her savings account at the end of the 3 years.
Give your answer correct to the nearest yen.

.... yen

(Total for Question is 3 marks)

60. 4MA1/1HR_Winter_2021_Q4 1 point

A train journey from Paris to Amsterdam took 3 hours 24 minutes.
The total distance the train travelled was 433.5 km.

Work out the average speed of the train.
Give your answer in kilometres per hour.

.... km/h

(Total for Question is 3 marks)

61. 4MA1/1HR_Winter_2021_Q1 1 point

Show that $3\frac{1}{5} \times 1\frac{5}{6} = 5\frac{13}{15}$

(Total for Question is 3 marks)

62. 4MA1/1H_Winter_2021_Q15 1 point

$$a = \frac{v - u}{t}$$

$v = 9.6$ correct to 1 decimal place
$u = 3.8$ correct to 1 decimal place
$t = 1.84$ correct to 2 decimal places

Calculate the upper bound for the value of a.
Give your answer as a decimal correct to 2 decimal places.
Show your working clearly.

....

(Total for Question is 3 marks)

63. 4MA1/1H_Winter_2021_Q7 1 point

Chen invests 40 000 yuan in a fixed-term bond for 3 years.

The fixed-term bond pays compound interest at a rate of 3.5% each year.

(a) Work out the value of Chen's investment at the end of 3 years.
Give your answer to the nearest yuan.

.... yuan
(3)

Wang invested P yuan.
The value of his investment decreased by 6.5% each year.

At the end of the first year, the value of Wang's investment was 30 481 yuan.

(b) Work out the value of P.

$P =$ ....
(3)

64. 4MA1/1H_Winter_2021_Q6

1 point

(a) Write 2 840 000 000 in standard form.

....

(1)

(b) Write 2.5×10^{-4} as an ordinary number.

....

(1)

Pieter owns a currency conversion shop.

Last Monday, Pieter changed a total of 20 160 rand into a number of different currencies.

He changed $\frac{3}{10}$ of the 20 160 rand into euros.

He changed the rest of the rands into dollars, rupees and francs in the ratios 9 : 5 : 2

Pieter changed more rands into dollars than he changed into francs.

Work out how many more.

.... rand

(Total for Question is 4 marks)

The sum of the first N terms of an arithmetic series, S, is 292
The 2nd term of S is 8.5
The 5th term of S is 13

Find the value of N.
Show clear algebraic working.

$N =$ ....

5 marks

67. 4MA1/1H_Summer_2021_Q20

1 point

$$P = \frac{t - w}{y}$$

t = 9.7 correct to 1 decimal place

w = 5.9 correct to 1 decimal place

y = 3 correct to 1 significant figure

Calculate the upper bound for the value of P.
Show your working clearly.

....

3 marks

68. 4MA1/1H_Summer_2021_Q17 1 point

Using algebra, prove that, given any 3 consecutive even numbers, the difference between the square of the largest number and the square of the smallest number is always 8 times the middle number.

3 marks

69. 4MA1/1H_Summer_2021_Q16 1 point

Use algebra to show that the recurring decimal $0.28\dot{1}\dot{3} = \dfrac{557}{1980}$

2 marks

70. 4MA1/1H_Summer_2021_Q10 1 point

Jethro has sat 5 tests.
Each test was marked out of 100 and Jethro's mean mark for the 5 tests is 74

Jethro has to sit one more test that is also to be marked out of 100

Jethro wants his mean mark for all 6 tests to be at least 77

Work out the least mark that Jethro needs to get for the last test.

....

3 marks

71. 4MA1/1H_Summer_2021_Q8 1 point

The table shows the populations of five countries.

Country	Population
China	1.4×10^9
Germany	8.2×10^7
Sweden	9.9×10^6
Fiji	9.1×10^5
Malta	4.3×10^5

(a) Work out the difference between the population of China and the population of Germany.
Give your answer in standard form.

....
(2)

Given that

$$\text{population of Fiji} = \frac{1}{k} \times \text{population of Sweden}$$

(b) work out the value of k.
Give your answer correct to the nearest whole number.

$k =$ ....
(2)

72. 4MA1/1H_Summer_2021_Q7

1 point

$-4 \leqslant 2y < 6$

y is an integer.

(a) Write down all the possible values of y.

....

(2)

(b) Solve the inequality $7t - 3 \leqslant 2t + 31$

Show your working clearly.

....

(2)

73. 4MA1/1H_Summer_2021_Q4 1 point

On a farm there are chickens, ducks and pigs.

The ratio of the number of chickens to the number of ducks is $7:2$
The ratio of the number of ducks to the number of pigs is $5:9$
There are 36 pigs on the farm.

Work out the number of chickens on the farm.

....

3 marks

74. 4MA1/1H_Summer_2021_Q1 1 point

A plane flew from Madrid to Dubai.

The distance the plane flew was 5658 km.
The flight time was 8 hours 12 minutes.

Work out the average speed of the plane.

.... km/h

3 marks

75. 4MA1/1HR_Winter_2020_Q13 1 point

(a) Use algebra to show that $0.5\dot{7}\dot{2} = \frac{63}{110}$

(2)

Given that y is a prime number,

(b) express $\frac{3}{2-\sqrt{y}}$ in the form $\frac{a+b\sqrt{y}}{c-y}$ where a, b and c are integers.

....

(2)